

# Solving Systems by Elimination with Scaling

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*Scaling one equation, scaling both equations, and building a system from a real situation before eliminating*

**The organizer.** Elimination needs one variable's coefficients to be opposites (or equal). When they are not, *scale*: multiply every term of an equation — both sides — by a constant. Use this layout on every problem.

| Step            | What you write                         | Why                                         |
|-----------------|----------------------------------------|---------------------------------------------|
| Target          | "eliminate $y$ "                       | pick the variable that is cheaper to match  |
| Scale           | multiplier next to each equation       | make the coefficients opposites             |
| Combine         | add the scaled equations               | the target variable drops out               |
| Back-substitute | put the value into either original     | find the second variable                    |
| Check           | test the pair in <i>both</i> originals | a pair that fits only one is not a solution |

*The order of work for every problem on this sheet. No values shown: use the numbers given in each problem.*

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## Directions.

- Write the multiplier you use beside each equation you scale.
- Solve for both variables and write the ordered pair on the answer line.
- Check your pair in both original equations before you write it down.

- 1.** A school store sells notebooks and binders. Two notebooks and one binder cost \$11. One notebook and three binders cost \$18. Let  $x$  be the price of a notebook and  $y$  the price of a binder, both in dollars.

$$2x + y = 11 \qquad x + 3y = 18$$

Scale one equation, eliminate a variable, and find both prices.

Answer: \_\_\_\_\_

- 2.** Solve by elimination. Scaling one equation is enough here.

$$4x - 3y = 5 \qquad x + y = 3$$

Answer: \_\_\_\_\_

**3.** A theater sold 12 tickets to one show for \$88 in all. Adult tickets cost \$9 each and child tickets cost \$5 each. Let  $a$  be the number of adult tickets and  $c$  the number of child tickets.

$$a + c = 12 \qquad 9a + 5c = 88$$

Solve the system by elimination and state how many of each ticket were sold.

Answer: \_\_\_\_\_

**4.** Solve by elimination. Neither pair of coefficients matches, so both equations must be scaled.

$$3x + 4y = 10 \qquad 5x + 6y = 16$$

Answer: \_\_\_\_\_

5. Solve by elimination.

$$5x + 2y = 4$$

$$3x + y = 3$$

Answer: \_\_\_\_\_

6. A bakery sold 50 items on Saturday and took in \$172. Muffins sell for \$3 each and scones for \$4 each. Let  $m$  be the number of muffins and  $s$  the number of scones.

$$m + s = 50$$

$$3m + 4s = 172$$

Solve the system by elimination and state how many of each were sold.

Answer: \_\_\_\_\_

- 7.** Solve by elimination. Choose the variable that needs the smaller multipliers, and say which one you chose.

$$4x + 5y = 7$$

$$6x + 7y = 11$$

Answer: \_\_\_\_\_

- 8.** A canoe travels 24 kilometres downstream in 2 hours and the same 24 kilometres back upstream in 3 hours. Let  $b$  be the canoe's speed in still water and  $c$  the speed of the current, both in kilometres per hour. Downstream the speeds add, upstream they subtract:

$$2b + 2c = 24$$

$$3b - 3c = 24$$

Solve the system by elimination and state both speeds in kilometres per hour.

Answer: \_\_\_\_\_

**9. Find and fix the mistake.** Maya is solving

$$x + 4y = 14 \qquad 3x - 2y = 14$$

She writes: “Multiply the first equation by 3 to match the  $3x$ : that gives  $3x + 12y = 14$ .”

- (a) Explain in one or two sentences what is wrong with her scaled equation and why multiplying *every* term by 3 leaves the solution unchanged.
- (b) Scale correctly and solve the system.

Answer: \_\_\_\_\_

**10.** A lab technician mixes a 30% salt solution with a 50% salt solution to make 20 litres of a 45% salt solution. Let  $x$  be the litres of the 30% solution and  $y$  the litres of the 50% solution. The volumes add, and the salt adds:

$$x + y = 20 \qquad 30x + 50y = 45 \cdot 20$$

Solve the system by elimination and state how many litres of each solution are used.

Answer: \_\_\_\_\_